

Supplemental Material for “A Posteriori Error Certificates (APECs): Certifying Uncertainty Through the Soft-Mode Bottleneck in BaTiO₃”

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This Supplemental Material provides extended formalism (Sections IA–IE), the certificate schema specification (Section II), supporting data tables (Section III), the complete BaTiO₃ literature library (Section IV), certificate data (Section V), and the extraction case study (Section VI). It also records the independent SCF convergence campaign used to check the cutoff and k -mesh behavior of the 40-atom BaTiO₃ supercell and the leave-one-out calibration check used to test the certificate uncertainties. Schematic downstream process-chain and residual-diagnostic examples are included to clarify how the schema can be extended without treating those extensions as completed main-text results.

I. S1. EXTENDED FORMALISM

A. S1.1. Diagonal and Covariance-Aware Fusion

The main text uses two related covariance approximations. First, within a single certificate, the reported convergence, methodological, intrinsic DFT, and model-form components are combined with a diagonal covariance approximation unless material-specific covariance data are available. Second, when multiple certified entries are fused, the default inverse-variance weighted (IVW) estimator assumes that the entries are exchangeable and conditionally independent after phase, tensor-component, and target-observable partitioning.

For observations collected in a vector $\mathbf{y} = (P_1, \dots, P_n)^\top$ with covariance matrix Σ , the generalized least-squares fused estimate is

$$\hat{P}_{\text{GLS}} = \frac{\mathbf{1}^\top \Sigma^{-1} \mathbf{y}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad \sigma_{\text{GLS}}^2 = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}. \quad (\text{S1})$$

If Σ is diagonal, this reduces to the IVW rule used in the BaTiO₃ case study. If source-level correlations are known or can be estimated, they should enter as off-diagonal terms. Examples include shared pseudopotentials, shared training/calibration sets, copied benchmark values, identical functional families, or multiple records generated by the same workflow.

When correlations are suspected but not yet quantified, the conservative certificate behavior is to partition more aggressively and avoid claiming spurious precision. A useful diagnostic is the effective sample size

$$N_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_{ij} w_i w_j R_{ij}}, \quad (\text{S2})$$

where w_i are scalar fusion weights and R_{ij} is the correlation matrix. For independent entries, R is the identity and N_{eff} approaches the usual Kish effective sample size; for strongly correlated entries, N_{eff} can be much smaller than the count of records. This is the formal version of the main-text caution that literature and database fusion should not treat every row as a new independent experiment.

B. S1.2. Hierarchical Bayesian Fusion

The inverse-variance weighting and bias-corrected fusion methods described in the main text assume that systematic biases are either negligible or known. When biases are uncertain or material-dependent, hierarchical modeling treats them as draws from a population distribution [1, 2]:

$$P_i | P_{\text{true}}, \beta_i \sim \mathcal{N}(P_{\text{true}} + \beta_i, \sigma_i^2) \quad (\text{S3})$$

$$\beta_i \sim \mathcal{N}(\mu_\beta, \tau_\beta^2) \quad (\text{S4})$$

$$P_{\text{true}} \sim \mathcal{N}(\mu_0, \sigma_0^2) \quad (\text{S5})$$

where P_i is the i -th estimate, β_i its functional-specific bias, σ_i its reported uncertainty, and (μ_β, τ_β) are hyperparameters governing the bias distribution. The prior on P_{true} can be uninformative ($\sigma_0 \rightarrow \infty$).

The posterior $p(P_{\text{true}}|\{P_i\})$ can be computed analytically for this conjugate Gaussian model. When the bias hyperparameters are unknown, they can be estimated from the data (empirical Bayes) or sampled via MCMC for fully Bayesian inference.

This approach is particularly valuable when fusing data across functionals (LDA, PBE, hybrid) that have known systematic tendencies: LDA overbinds, PBE underbinds, and the bias magnitude varies by material class. The hierarchical model automatically downweights estimates whose bias is large relative to the population, without requiring prior knowledge of individual biases.

The main text exercises the simpler inverse-variance weighting (appropriate when biases are small relative to random uncertainties). The hierarchical model becomes important when fusing across functionals with strongly different systematic tendencies—a regime that arises naturally as the certificate framework scales to multi-material, multi-functional databases.

C. S1.3. Monte Carlo Propagation

For derived quantities $Q = f(P_1, P_2, \dots, P_n)$ where the linear propagation formula in the main text breaks down due to nonlinearity or large relative uncertainties, Monte Carlo sampling provides robust estimates:

$$Q^{(k)} = f(P_1^{(k)}, \dots, P_n^{(k)}), \quad P_i^{(k)} \sim \mathcal{N}(\hat{P}_i, \sigma_{P_i}^2) \quad (\text{S6})$$

for $k = 1, \dots, K$ samples. The distribution of $\{Q^{(k)}\}$ provides the full posterior, from which mean, standard deviation, percentiles, and skewness can be extracted.

This approach is consistent with the Monte Carlo method for uncertainty evaluation recommended by the Guide to the Expression of Uncertainty in Measurement [3–5]. For the BaTiO₃ case study, the soft-mode model $r_{33} \propto Z^{*2}/(\varepsilon\omega^2)$ is sufficiently nonlinear (with $\gamma = \pm 2$ exponents) that Monte Carlo propagation would capture asymmetric uncertainty distributions—e.g., when ω_{soft} is near the phase transition and its distribution is skewed. The main text uses linear propagation, which is adequate for the current parameter regime (3–8% input uncertainties). Monte Carlo propagation is the schema-supported route for certificates outside this linear regime, including cases where ω_{soft} approaches zero and relative uncertainty summaries become ill-conditioned.

D. S1.4. Multi-Stage Process Chains

Real computational workflows involve sequential stages, each with its own certificate:

$$\boxed{\text{DFT}} \xrightarrow{c_1} \boxed{\text{Property}} \xrightarrow{c_2} \boxed{\text{Model}} \xrightarrow{c_3} \boxed{\text{Prediction}} \quad (\text{S7})$$

Uncertainties compound through the chain as

$$\sigma_{\text{final}}^2 = \sigma_1^2 \cdot G_1^2 + \sigma_2^2 \cdot G_2^2 + \sigma_3^2 \quad (\text{S8})$$

where $G_i = \partial(\text{stage } i+1)/\partial(\text{stage } i)$ are sensitivity (gain) factors. This formulation makes explicit where in a pipeline uncertainty reduction efforts will have the greatest impact. The main text demonstrates stages 1–2 (DFT $\rightarrow$ property $\rightarrow$ derived r_{33}) and discusses how the same certificate decomposition can be propagated to device metrics such as $V_\pi L$ and switching energy.

For electro-optic modulator design, the certified bulk or single-domain Pockels coefficient is not the end of the uncertainty chain. The response must pass through additional modeling stages such as domain averaging, waveguide geometry, optical–RF overlap, propagation loss, and fabrication tolerances. For example, the common half-wave voltage–length product is

$$V_\pi L = \frac{\lambda d}{2n^3 r_{\text{eff}} \Gamma}, \quad (\text{S9})$$

where λ is wavelength, d is electrode gap, n is refractive index, r_{eff} is the effective Pockels coefficient, and Γ is the optical–RF field overlap [6–8]. Each of these terms is a candidate certificate target once the corresponding microstructure, optical-mode, and fabrication models are specified.

The following schematic chain is therefore a schema demonstration, not a completed device calculation. The DFT and Property stages use the uncertainty decomposition developed in the main text; the Effective and Device stages are schematic estimates showing where additional certified domain, strain, geometry, overlap, and loss models would enter if their exchangeability classes and model discrepancies were specified.

Illustrative multi-stage uncertainty propagation from DFT to downstream response-model priors.

Stage	Quantity	Value	Rel. uncertainty
DFT	ω_{soft}	172 cm ⁻¹	10%
↓ (×2 amplification)			
Property (single calc)	$r_{33} (zz)$	94 pm/V	28%
Property (library-fused)	$r_{33} (zz)$	94 pm/V	24%
<i>Downstream stages to be certified:</i>			
↓ (microstructure averaging)			
Effective	r_{eff}	~40 pm/V	~26%
↓ (device model)			
Device	$V_{\pi}L$	~2.5 V·cm	~27%

The current propagation chain does not capture optical propagation loss, which Kim and Demkov [9] have shown can be dominated by non-absorptive scattering at ferroelectric domain walls in BaTiO₃ waveguides. Loss is therefore a separate out-of-scope target observable rather than a contribution that can be folded into r_{eff} or $V_{\pi}L$ without its own certificate.

E. S1.5. Convergence Uncertainty Estimators

The main text defines the total convergence uncertainty as the quadrature sum $\sigma_{\text{conv}}^2 = \sigma_{E_{\text{cut}}}^2 + \sigma_k^2 + \sigma_{\text{SCF}}^2 + \dots$. Here we provide the detailed estimators for each component.

For the plane-wave cutoff, the uncertainty combines the estimated truncation error at the working cutoff with the sensitivity of the property to cutoff changes:

$$\sigma_{E_{\text{cut}}}^2 = \left(\frac{\partial P}{\partial E_{\text{cut}}} \right)^2 \cdot \delta E^2 + \epsilon_{E_{\text{cut}}}^2(E_{\text{used}}) \quad (\text{S10})$$

where δE is the step size in the convergence test and $\epsilon_{E_{\text{cut}}}(E) = A_E \exp(-E/E_0)$ is the parametric truncation error model.

For k -point sampling, the uncertainty includes a term for anisotropic sampling effects:

$$\sigma_k^2 = \epsilon_k^2(N_k) + \sigma_{\text{symmetry}}^2 \quad (\text{S11})$$

where $\epsilon_k(N_k) = B/N_k^\alpha$ and σ_{symmetry} captures the spread in equivalent-by-symmetry quantities (e.g., σ_{ii} tensor components in a nominally cubic system).

The self-consistency contribution is estimated from the sensitivity of the property to the residual density at the final SCF iteration:

$$\sigma_{\text{SCF}} \approx \left| \frac{\partial P}{\partial \rho} \right| \cdot \delta \rho_{\text{final}} \quad (\text{S12})$$

where $\delta \rho_{\text{final}}$ is the SCF residual norm.

Residual-Based Diagnostic Indicators

Several quantities available from a single calculation can provide qualitative convergence flags when explicit convergence sweeps are unavailable. These indicators are not used as calibrated uncertainty contributions in the main BaTiO₃ certificate; they are optional diagnostic metadata for automated certificate generation.

The decay rate of SCF residuals,

$$r_{\text{SCF}} = \frac{\log(\delta \rho_1 / \delta \rho_N)}{N}, \quad (\text{S13})$$

indicates electronic-structure conditioning. Slow convergence, for example $r_{\text{SCF}} < 0.5$, suggests a difficult electronic structure and may motivate a conservative tier downgrade when no explicit convergence data are available.

The spread of stress tensor eigenvalues in nominally symmetric systems,

$$\Delta\sigma = \max_i(\sigma_{ii}) - \min_i(\sigma_{ii}), \quad (\text{S14})$$

can flag anisotropic sampling or incomplete relaxation. Non-zero forces at equilibrium positions similarly indicate incomplete relaxation, basis-set incompleteness, or a mismatch between the claimed symmetry and the actual calculated structure.

II. S2. CERTIFICATE SCHEMA SPECIFICATION

A. S2.1. JSON Schema

The certificate schema is defined as a JSON-serializable structure. A compact example for a typed BaTiO₃ optical-gap certificate is:

```
{
  "property": {
    "name": "band_gap",
    "target_observable": "optical_absorption_onset",
    "value": 3.08,
    "unit": "eV"
  },
  "uncertainties": {
    "convergence": 0.096,
    "methodological": 0.462,
    "intrinsic_dft": 0.246,
    "model_form": 0.246,
    "total": 0.585,
    "relative_total": 0.190
  },
  "quality_tier": "RED",
  "exchangeability_key": {
    "material": "BaTiO3",
    "phase": "tetragonal",
    "target_observable": "optical_absorption_onset",
    "temperature_regime": "room_temperature"
  },
  "provenance": {
    "evidence_set": "literature_proxy_plus_optical_anchor",
    "included_targets": ["TB-mBJ_gap", "HSE06_gap",
                        "optical_absorption_onset"],
    "excluded_targets": ["LDA_Kohn-Sham_gap", "PBE_Kohn-Sham_gap"],
    "exclusion_reason": "non_exchangeable_target_observable"
  },
  "fusion": {
    "method": "typed_inverse_variance",
    "covariance_policy": "diagonal_until_source_correlations_available",
    "effective_sample_size": null
  },
  "descriptor_selection": {
    "mode_identification": "not_applicable",
    "selected_descriptor": "optical_absorption_onset"
  },
  "model_discrepancy": {
    "discrepancy_provenance": "referenced",
```

```

    "carried_magnitude": 0.246,
    "status": "carried_not_corrected"
  }
}

```

Key fields:

- `property.target_observable`: the physical or computational quantity being certified; this prevents semilocal Kohn–Sham gaps, quasiparticle gaps, and optical onsets from being fused accidentally.
- `uncertainties.model_form`: nonzero for derived properties or calibrated proxy observables (e.g., r_{33} via the soft-mode model or a hybrid-functional band-gap calibration).
- `exchangeability_key`: the minimal fields that must match before entries can be fused.
- `provenance.phase`: crystallographic phase (`cubic`, `tetragonal`, etc.).
- `provenance.tensor_component`: for anisotropic properties (`zz`, `xx`, `isotropic`, or `null`).
- `descriptor_selection`: the descriptor substituted into any calibrated derived expression, including mode-identification status for phonon-mediated response properties.
- `model_discrepancy.discrepancy_provenance`: whether the model discrepancy is referenced from literature or computed in the same workflow, and whether it is carried as uncertainty or used as a correction.
- `fusion.effective_sample_size`: the correlation-aware number of exchangeable sources used in a fused posterior, together with the stated `covariance_policy`.

The schema captures all information needed to reconstruct uncertainty estimates and enables downstream consumers to re-derive tier assignments under alternative threshold conventions.

B. S2.2. Support-Repository Extraction Interface

The paper support repository provides self-contained scripts that exercise the certificate extraction and aggregation logic without requiring any external development package. The extraction case study reads the curated JSON library, partitions entries by phase and tensor component, computes inverse-variance weighted averages where exchangeability is justified, and writes an audit summary:

```
python scripts/extraction_case_study.py
```

The generated summary records source classes, property counts, weighted tetragonal averages, phase/tensor r_{33} checks, experimental calibration targets, and the policy used to separate curated fusion inputs from extraction stress tests. The machine-readable inputs are `data/bto_literature_library.json` and `data/posterior_certificates.json`; the principal output is `outputs/extraction_case_study_summary.json`.

C. S2.3. Retrospective Database Extraction and Fallback Hierarchy

The quantitative BaTiO₃ electro-optic fusion in the main text is driven by curated literature entries and the production Quantum ESPRESSO calculation. Public database records are used as extraction stress tests and as examples of how the schema behaves when provenance is incomplete. The extraction hierarchy is:

1. **Complete provenance**: use explicit calculation settings, functional, pseudopotential, structure, phase, tensor component, and convergence metadata. Assign convergence uncertainty from reported convergence studies where available.
2. **Partial provenance**: retain the record but estimate missing convergence fields from class-level surrogate or heuristic models. Mark the certificate as reconstructed and downgrade the quality tier if the missing field affects the target observable.

3. **Ambiguous target observable:** retain the record in the audit trail but exclude it from fusion until the target can be typed. The band-gap case study is the canonical example: semilocal Kohn–Sham gaps are not fused with optical absorption onsets without a calibrated reduction map.
4. **Phase or tensor mismatch:** partition rather than average. Cubic and tetragonal BaTiO₃ entries, or *zz* and isotropic tensor conventions, are kept in separate exchangeability classes.
5. **Insufficient provenance:** store the entry as Red-tier evidence for auditability, but do not use it in quantitative fusion unless a conservative uncertainty large enough to cover the ambiguity can be justified.

This hierarchy is intentionally conservative. It allows retrospective certification of database-scale data without pretending that missing metadata are harmless, and it explains why Materials Project and Kyoto PhononDB records are useful demonstrations of extraction behavior without dominating the phase-filtered electro-optic estimate in the main paper.

Role of each data source in the BaTiO₃ certificate workflow. The final electro-optic posterior uses the curated literature library and the production Quantum ESPRESSO calculation; public database records are retained as retrospective extraction and metadata-stress tests unless their target observable, phase, and tensor convention are sufficiently typed for fusion.

Source class	Example records	Role in this work
Curated DFPT literature	Ghosez, Zhong, Veithen, Tinte	Quantitative fusion input
Production calculation	QE 7.5 PBE/PAW, $2 \times 2 \times 2$ BTO	Certificate generation input
PhononDB	PBEsol phonon/lattice entries	Typed literature/database support
Materials Project	mp-5986, mp-2998	Extraction stress test; not final fusion driver
Experimental references	Zgonik, Reynaud, Suceava	Calibration/check targets, not DFT entries

III. S3. SUPPORTING DATA TABLES

A. S3.1. Functional Bias Statistics

TABLE S1. Functional bias statistics for common properties, compiled from literature benchmarks. LC154: 154-compound lattice constant set [10]; SC40: 40-semiconductor band gap set [11]; FERE: fitted elemental reference energies [12].

Property	Functional	Mean bias	MAE	Ref. set
Lattice const. (%)	LDA	−1.8	2.1	LC154
	PBE	+1.2	1.4	LC154
	PBEsol	+0.2	0.5	LC154
	SCAN	+0.1	0.4	LC154
Band gap (%)	LDA	−42	42	SC40
	PBE	−38	38	SC40
	HSE06	−8	15	SC40
	G_0W_0 @PBE	+3	12	SC40
ΔH_f (meV/at)	PBE	+85	120	FERE
	SCAN	+25	65	FERE
	r ² SCAN	+15	50	FERE

B. S3.2. Band-Gap Certificate Case Study

Band-gap entries require additional typing beyond the scalar value and uncertainty shown in Table S1. A semilocal Kohn–Sham eigenvalue gap, a hybrid-functional gap calibrated by the fraction of exact exchange, a quasiparticle gap, and an optical absorption onset are different target observables. Consequently, band-gap data should be partitioned by target type before fusion. Prior HSE06 calibration against empirical BaTiO₃ gap ranges can be represented as a

model-form layer: the exact-exchange mixing parameter, calibration dataset, accepted empirical range, and residual calibration uncertainty become certificate metadata rather than implicit choices.

We implement this as a formal certificate case study rather than a prose-only example. The evidence table contains LDA and PBE Kohn–Sham gaps, TB-mBJ and HSE06 proxy physical-gap records, and a room-temperature optical absorption anchor [13]. The semilocal Kohn–Sham entries are retained in the audit trail but excluded from optical-gap fusion unless a calibrated reduction map is supplied. The generated typed optical-gap certificate gives $E_g = 3.08$ eV with $\sigma_{\text{total}}/|E_g| = 19.0\%$, placing it in the Red tier even though the experimental optical anchor alone would be Green. This is the desired behavior: the certificate reports uncertainty in using a heterogeneous DFT/proxy evidence set for a physical optical target, not merely the precision of one measurement.

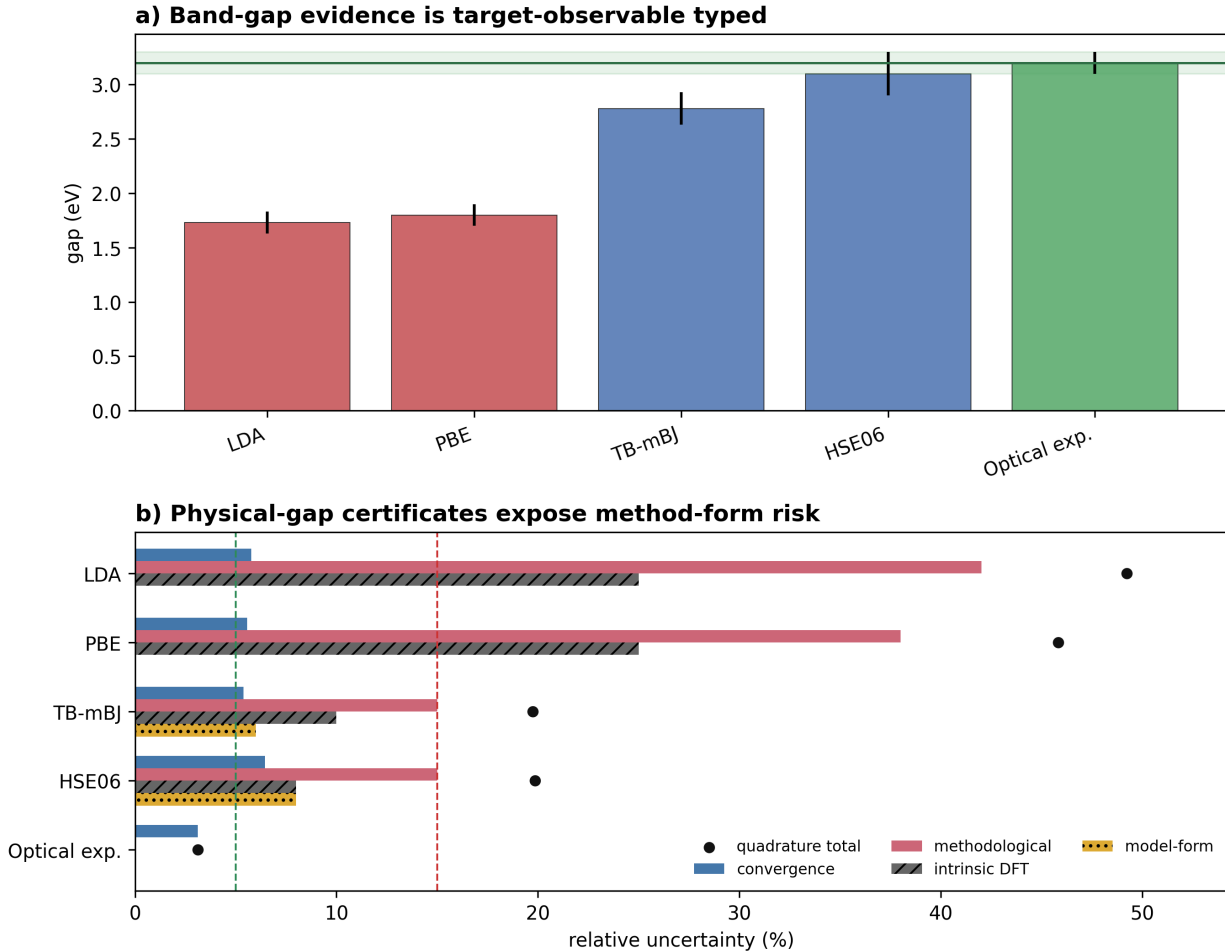


FIG. S1. Band-gap certificate case study. (a) LDA and PBE Kohn–Sham gaps are well-defined computational outputs but sit far below the room-temperature optical absorption target. TB-mBJ and HSE06 are closer physical-gap proxies, while the empirical optical reference defines the target scale. (b) The uncertainty decomposition keeps semilocal Kohn–Sham gaps as Red-tier evidence and reports the promoted typed optical-gap certificate as Red until the HSE06 calibration study is imported as an explicit model-form layer.

The machine-readable artifacts are `outputs/band_gap_certificate/band_gap_certificate_case_study.json` and `outputs/band_gap_certificate/band_gap_certificate_case_study.csv`. The main certificate file also includes the promoted `E_gap_optical-typed` record, while preserving the original production DFPT electro-optic certificates as a distinct 12-record subset.

TABLE S2. Soft-mode frequency in tetragonal BaTiO₃ from published DFPT calculations. The cross-functional spread ($\sigma \approx 5 \text{ cm}^{-1}$, or $\sim 3\%$) drives the methodological uncertainty component for ω_{soft} .

Functional	$\omega_{\text{soft}} \text{ (cm}^{-1}\text{)}$	Phase	Source
LDA	180	Tetragonal	Ghosez [14]
LDA	175	Tetragonal	Zhong [15]
PBE	185	Tetragonal	Veithen [16]
PBE	172	Tetragonal	This work
PBEsol	178	Tetragonal	PhononDB [17]
Experiment	~ 180	Tetragonal	Lines & Glass [18]

C. S3.3. Cross-Functional Soft-Mode Comparison

D. S3.4. Convergence Model Parameters

TABLE S3. Preliminary convergence model parameters (oxide perovskite class). Values are order-of-magnitude estimates informed by published convergence behavior [14, 19]; they have not been systematically fitted and should not be used quantitatively without validation. A and B are prefactors for the cutoff and k -point models defined in the main text (Eqs. 20–21), E_0 is the cutoff decay constant, and α is the k -point power law exponent. These parameters are not used as fitted soft-mode curve coefficients in Fig. 1 of the main text; that figure is a heuristic visual interpolant based on Table S4.

Property	A	$E_0 \text{ (eV)}$	B	α
Total energy (meV/at)	150	120	85	1.2
Forces (meV/Å)	80	100	45	1.5
Lattice constant (%)	0.8	90	0.4	1.8
Bulk modulus (%)	5.2	110	3.1	1.4
Band gap (%)	3.5	150	2.8	1.6

E. S3.5. Heuristic Convergence Estimates

TABLE S4. Heuristic convergence uncertainties for perovskite DFPT calculations when convergence data is unavailable. These are expert estimates informed by Refs. [14, 19], not fitted parameters.

Property	$E_{\text{cut}} \geq 60 \text{ Ry}$	40–60 Ry	$< 40 \text{ Ry}$
Z^*	1–2%	2–3%	3–5%
ϵ^∞	1.5–3%	3–5%	5–8%
ω_{soft}	3–5%	5–8%	8–12%
Lattice parameters	0.2–0.4%	0.4–0.8%	0.8–1.5%

F. S3.6. Independent SCF Convergence Campaign

The main text includes an independent Quantum ESPRESSO SCF convergence study performed on the 40-atom tetragonal BaTiO₃ supercell. The raw outputs are distributed with the manuscript support files. Each filename has the form {functional}_{ecut}_{kmesh}.scf.out. The study contains 28 PBE calculations, with cutoffs from 30 to 90 Ry and k meshes from 4^3 to 10^3 , and 9 PBEsol calculations, with cutoffs from 40 to 80 Ry and k meshes from 4^3 to 8^3 . All 37 calculations reached SCF convergence and normal job termination.

Summary of the independent SCF total-energy convergence campaign. Errors are measured relative to the highest cutoff and densest k mesh available within each functional. These data certify the

total-energy and ground-state numerical scale; they do not replace DFPT convergence tests for soft phonons or electro-optic response tensors.

Functional	Runs	Reference	60 Ry, 6 ³ error	Highest-cutoff k spread
PBE	28	90 Ry, 10 ³	11.35 meV/atom	5.7×10^{-4} meV/atom
PBEsol	9	80 Ry, 8 ³	10.51 meV/atom	1.3×10^{-3} meV/atom

For both functionals the cutoff dominates the total-energy convergence. The sampled k -mesh dependence is already negligible at the highest cutoff for the 40-atom cell, whereas the 60 Ry grid remains about 10–11 meV/atom above the internal high-cutoff reference. This supports treating total-energy convergence as a cutoff-controlled contribution while retaining separate, more conservative uncertainty estimates for soft-mode and response-property convergence.

The processing script also serializes the 37 SCF calculations as diagnostic total-energy convergence certificates. These records attach absolute convergence uncertainty, SCF residual uncertainty, provenance metadata, and a ground-state-energy tier to each cutoff/ k -mesh calculation. They are intended for convergence-model calibration and auditability of the SCF campaign, not for inverse-variance fusion across cutoff grids as independent estimates of a final physical property. The serialized records are provided in `scf_energy_certificates.json` and `scf_energy_certificates.csv` with the processed convergence outputs.

G. S3.7. Leave-One-Out Calibration Check

The main text reports a calibration check on the phase/component-consistent subsets of the BaTiO₃ literature library. For every subset with at least three computational entries, one entry is held out, the remaining entries are fused by inverse-variance weighting, and the held-out residual is standardized as

$$z_i = \frac{P_i - \hat{P}_{-i}}{\sqrt{\sigma_i^2 + \sigma_{-i}^2}}, \quad (\text{S15})$$

where $\hat{P}_{-i}$ and σ_{-i} are the leave-one-out fused mean and uncertainty. This denominator treats both the held-out entry and the fused prediction as uncertain. If the reported certificate uncertainties are calibrated and the entries are approximately exchangeable within each subset, the pooled z_i values should be distributed approximately as $\mathcal{N}(0, 1)$.

Leave-one-out calibration summary for the BaTiO₃ certificate library. The check is intentionally limited to computational subsets with at least three entries; it is a smoke test for gross over- or under-confidence, not a replacement for a large benchmark campaign. We therefore treat it as an Amber-level meta-certificate on the present library’s calibration status.

Statistic	Observed	Reference
Leave-one-out residuals	19	—
Phase/component groups	5	—
Mean z	−0.03	0
RMS z	0.99	1
Pooled reduced χ^2	0.99	1
Coverage, $ z < 1$	68%	68%
Coverage, $ z < 1.96$	95%	95%

The five tested groups are P_s in the tetragonal scalar/isotropic subset, Z_{T1}^* in the cubic-reference isotropic subset, ε^∞ in the tetragonal scalar/isotropic subset, tetragonal lattice parameter a , and tetragonal soft-mode frequency. Table S5 shows that the pooled statistics are close to the standard-normal expectation, with RMS $z = 0.99$ and reduced $\chi^2 = 0.99$, while the per-subset RMS values bracket unity from both sides. The room-temperature bulk anchored-consistency check for the library-fused isotropic r_{33} gives $z = 0.11$ relative to the Zgonik et al. [20] reference value, using the combined library and experimental uncertainties. Because the soft-mode expression is anchored to this bulk reference, this comparison is an internal consistency check rather than an independent validation of the absolute electro-optic scale. These results support the claim that the present certificate library is not grossly miscalibrated at the scale probed by available data, but they do not certify transferability across broader material classes.

TABLE S5. Per-subset leave-one-out calibration of the BaTiO₃ input library. Each subset is a (property, phase, tensor-component) group containing at least three exchangeable computational entries. The z -scores are standardized residuals against the inverse-variance leave-one-out prediction, $z_i = (P_i - P_{\text{LOO}})/\sqrt{\sigma_i^2 + \sigma_{\text{LOO}}^2}$. The pooled RMS lies within one sampling spread ($1/\sqrt{2n} \approx 0.16$) of unity, but the per-subset values bracket it from both sides, indicating mild bidirectional rather than uniform miscalibration.

Subset	n	mean z	RMS z	max $ z $
ω_{soft} , tet., iso.	5	+0.11	0.65	1.04
Z_{Ti}^* , cubic, iso.	4	-0.05	0.62	0.93
ε^∞ , tet., iso.	3	-0.03	0.87	1.09
P_s , tet., iso.	3	+0.08	1.37	1.76
a , tet., iso.	4	-0.25	1.35	2.50
Pooled	19	-0.03	0.99	2.50

IV. S4. COMPLETE BATiO₃ LITERATURE LIBRARY

Table S6 lists the BaTiO₃ posterior certificate library, with phase and tensor-component metadata for each entry, together with derived tensor-provenance rows needed to reproduce the final phase-filtered averages. This compilation spans DFPT studies from 1994 to 2025 and forms the basis for the weighted averages and phase-filtered analysis in the main text.

Full DOIs for all entries are provided in the machine-readable library file (`bto_literature_library.json`) in the repository.

V. S5. CERTIFICATE DATA

A. S5.1. Complete Certificates

Table S7 lists the four-component uncertainty decomposition for the 12 posterior certificates from the BaTiO₃ $2 \times 2 \times 2$ supercell DFPT calculation (PBE/PAW, 60 Ry cutoff, 6^3 k -points, Quantum ESPRESSO 7.5), plus the promoted typed band-gap case-study certificate described in Section IIIB.

B. S5.2. Soft-Mode Model-Form Arithmetic

The Red-tier r_{33} certificates include a nonzero model-form contribution because the electro-optic response is not computed as a full *ab initio* electro-optic tensor in the production workflow. Instead, it is inferred using the soft-mode expression

$$r_{33} = r_{\text{ref}} \left(\frac{\omega_{\text{ref}}}{\omega} \right)^2 \times \left(\frac{Z^*}{Z_{\text{ref}}^*} \right)^2 \times \left(\frac{\varepsilon_{\text{ref}}}{\varepsilon^\infty} \right).$$

Following the mode-decomposition comparison of Veithen et al. [16], we assign a 15% soft-mode approximation uncertainty for BaTiO₃. The production certificate also assigns a 9.8% reference calibration uncertainty to the fixed literature anchors ($r_{\text{ref}}, \omega_{\text{ref}}, Z_{\text{ref}}^*, \varepsilon_{\text{ref}}$) used in the calibrated expression. These two terms are combined in quadrature:

$$\sigma_{\text{model}} = \sqrt{(15.0\%)^2 + (9.8\%)^2} = 17.9\%. \quad (\text{S16})$$

This contribution is held fixed across the zz and isotropic conventions so that changes in the reported r_{33} values reflect the phase and tensor inputs rather than a change in the phenomenological model floor.

The methodological component of the derived r_{33} certificate is propagated from the input line items using the same Jacobian as the convergence and intrinsic-DFT components:

$$\sigma_{\text{meth}}(r_{33}) = \sqrt{(2 \times 8.0\%)^2 + (2 \times 3.0\%)^2 + (5.0\%)^2} = 17.8\%.$$

Together with $\sigma_{\text{conv}} = 6.5\%$, $\sigma_{\text{DFT}} = 11.2\%$, and $\sigma_{\text{model}} = 17.9\%$, this gives $\sigma_{\text{total}} = 28.4\%$ for both the zz and isotropic single calculation certificates.

TABLE S6. Complete BaTiO₃ literature library and tensor-provenance rows. Phase: C = cubic, T = tetragonal. Component: zz, xx, iso (isotropic average), or — (scalar). The cubic Born-charge rows include older cubic-reference or high-symmetry reference-structure records; tensor directions are equivalent by symmetry in that bin. Uncertainties are one standard deviation. Derived rows are included for auditability and should not be counted as independent source entries in inverse-variance fusion.

Parameter	Ph.	Comp.	Value	σ	Functional	Source
<i>Born effective charge Z_{Ti}^* (e)</i>						
Z_{Ti}^*	C	iso	7.16	0.40	LDA	Ghosez 1998
Z_{Ti}^*	C	iso	7.25	0.30	LDA	Zhong 1994
Z_{Ti}^*	C	iso	7.00	0.40	LDA	Ghosez 1998
Z_{Ti}^*	C	iso	7.50	0.30	PBE	Veithen 2005
Z_{Ti}^*	T	xx	5.70	0.20	LDA	Zhong 1994
Z_{Ti}^*	T	xx	7.17	0.20	PBE	This work
Z_{Ti}^*	T	zz	6.20	0.20	PBE	This work
Z_{Ti}^*	T	iso	6.85	0.20	PBE	This work (derived)
<i>Dielectric constant ϵ^∞</i>						
ϵ^∞	C	iso	6.90	0.30	LDA	Ghosez 1998
ϵ^∞	T	—	6.75	0.20	LDA	Veithen 2005
ϵ^∞	T	—	6.50	0.30	LDA	Ghosez 1998
ϵ^∞	T	—	7.00	0.25	PBE	Tinte [21]
ϵ^∞	T	zz	5.96	0.18	PBE	This work
<i>Soft-mode frequency ω_{soft} (cm⁻¹)</i>						
ω_{soft}	C	—	-95	20	LDA	Ghosez 1998
ω_{soft}	T	—	180	10	LDA	Ghosez 1998
ω_{soft}	T	—	175	15	LDA	Zhong 1995
ω_{soft}	T	—	185	12	PBE	Tinte [21]
ω_{soft}	T	—	178	8	PBEsol	PhononDB
ω_{soft}	T	—	172	5	PBE	This work
<i>Spontaneous polarization P_s (C/m²)</i>						
P_s	T	—	0.27	0.03	LDA	Zhong 1995
P_s	T	—	0.26	0.02	Expt	Merz 1953
P_s	T	—	0.26	0.01	PBEsol	Amounas 2024
P_s	T	—	0.30	0.02	PBE	Amounas 2024
<i>Lattice parameters ($\text{\AA}$)</i>						
a	T	—	3.940	0.020	LDA	Ghosez 1998
a	T	—	4.000	0.020	PBE	Tinte [21]
a	T	—	3.990	0.010	PBEsol	PhononDB
a	T	—	3.994	0.005	Expt	Kwei 1993
a	T	—	3.991	0.005	PBEsol	Amounas 2024
c	T	—	4.036	0.005	Expt	Kwei 1993
c	T	—	4.020	0.020	PBEsol	PhononDB
<i>Other</i>						
$Z_{\text{O},\parallel}^*$	T	zz	-5.71	0.30	LDA	Zhong 1994
$Z_{\text{O},\perp}^*$	T	xx	-2.11	0.20	LDA	Zhong 1994
Z_{Ba}^*	T	iso	2.74	0.10	LDA	Zhong 1994

C. S5.3. Phase-Correction Arithmetic

The main text shows five r_{33} values corresponding to different input conventions. The soft-mode model gives $r_{33} = r_{\text{ref}} \times (\omega_{\text{ref}}/\omega)^2 \times (Z^*/Z_{\text{ref}}^*)^2 \times (\epsilon_{\text{ref}}/\epsilon)$, with literature-standard reference values: $r_{\text{ref}} = 105 \text{ pm/V}$, $\omega_{\text{ref}} = 180 \text{ cm}^{-1}$, $Z_{\text{ref}}^* = 7.16 \text{ e}$, $\epsilon_{\text{ref}} = 6.5$.

- Unpartitioned library:** $\omega = 176$, $Z^* = 7.29$, $\epsilon = 6.51$.
 $r_{33} = 105 \times (180/176)^2 \times (7.29/7.16)^2 \times (6.5/6.51) = 114 \text{ pm/V}$.
- Tetragonal isotropic (library):** $\omega = 172$, $Z^* = 6.85$, $\epsilon = 6.31$.
 $r_{33} = 105 \times (180/172)^2 \times (6.85/7.16)^2 \times (6.5/6.31) = 108 \text{ pm/V}$.
- Tetragonal zz only (library):** $\omega = 172$, $Z^* = 6.20$, $\epsilon = 5.96$.
 $r_{33} = 105 \times (180/172)^2 \times (6.20/7.16)^2 \times (6.5/5.96) = 94 \text{ pm/V}$.
- Single calculation, zz convention:** $\omega = 172$, $Z^* = 6.20$ (zz), $\epsilon = 5.96$ (zz).
 $r_{33} = 94 \text{ pm/V}$ —matches the library zz row exactly.

TABLE S7. Complete four-component uncertainty decomposition for the 12 production DFPT certificates and the promoted typed band-gap certificate. All uncertainty components are relative (%). For the derived r_{33} , σ_{conv} , σ_{meth} , and σ_{DFT} are Jacobian-propagated from the input properties; σ_{model} captures the phenomenological soft-mode approximation. Both zz and isotropic tensor conventions are reported for r_{33} . The band-gap record is not a DFPT tensor output; it is a typed-observable case study.

Property	Value	σ_{conv}	σ_{meth}	σ_{DFT}	σ_{model}	σ_{total}	Tier
Z_{Ti}^* (mean) = 6.85 e		1.00	3.00	2.00	0	3.74	Green
$Z_{\text{Ti},zz}^* = 6.20 e$		1.00	3.00	2.00	0	3.74	Green
$Z_{\text{O},zz}^* = -5.05 e$		1.00	3.00	2.00	0	3.74	Green
Z_{Ba}^* (mean) = 2.75 e		1.00	3.00	1.00	0	3.32	Green
$\varepsilon_{xx}^\infty = 6.49$		1.50	5.00	3.00	0	6.02	Amber
$\varepsilon_{zz}^\infty = 5.96$		1.50	5.00	3.00	0	6.02	Amber
$\omega_{\text{soft}} = 172 \text{ cm}^{-1}$		3.00	8.00	5.00	0	9.90	Amber
$\omega_{\text{soft},z} = 144 \text{ cm}^{-1}$		3.00	8.00	5.00	0	9.90	Amber
$c/a = 1.011$		0.20	1.00	0.20	0	1.04	Green
$a = 3.993 \text{ \AA}$		0.20	1.00	0.20	0	1.04	Green
$r_{33} (zz) = 94 \text{ pm/V}$		6.50	17.8	11.2	17.9	28.4	Red
$r_{33} (\text{iso}) = 108 \text{ pm/V}$		6.50	17.8	11.2	17.9	28.4	Red
$E_g \text{ optical (typed)} = 3.08 \text{ eV}$		3.13	15.0	8.0	8.0	19.0	Red

5. **Single calculation, isotropic convention:** $\omega = 172$, $Z^* = 6.85$ (mean), $\varepsilon = 6.32$ (mean).
 $r_{33} = 108 \text{ pm/V}$ —matches the library isotropic row.

The present certificate uses $\omega_{\text{soft}} = 172 \text{ cm}^{-1}$ as the calibrated soft-mode descriptor for the bulk reference comparison. A separate raw DFPT entry, $\omega_{\text{soft},z} = 144 \text{ cm}^{-1}$, is retained in the certificate table as mode-provenance information rather than substituted into the calibrated r_{33} rows. If the 144 cm^{-1} mode were used in the zz arithmetic with the same Z^* and ε^∞ values, the central zz estimate would shift to approximately 134 pm/V . This sensitivity motivates the `mode_identification` and `descriptor_selection` fields in the certificate schema: the central value is conditional on the selected descriptor, and alternative descriptor forks can be retained as explicit audit metadata.

The consistency between single-calculation and library-fused values, when the tensor convention is held fixed, validates the extraction pipeline and demonstrates that the convention choice—not the number of sources—drives the spread in reported r_{33} values. The 14% spread between zz and isotropic conventions (94 vs 108 pm/V) quantifies the sensitivity to tensor averaging.

VI. S6. EXTRACTION CASE STUDY

A. S6.1. Pipeline Walkthrough

The extraction case study demonstrates how the certificate framework incorporates new literature data and traces the effect through the pipeline. The full runnable script (`extraction_case_study.py`) is available in the repository.

The workflow proceeds in six steps:

1. **Baseline:** Load the 25-entry literature library (1994–2024) and compute inverse-variance weighted averages for each parameter.
2. **Extract:** Parse six new entries from 2024–2025 publications: two P_s values from Amounas et al. [22] (PBEsol and PBE), one lattice constant, and three properties from the production DFPT calculation (ω_{soft} , Z_{Ti}^* , ε_{zz}^∞).
3. **Merge:** Add the new entries to the library, updating the total from 25 to 31.
4. **Recompute:** Recalculate inverse-variance weighted averages with the expanded library.
5. **Propagate:** Push the updated averages through the soft-mode model to obtain an updated r_{33} .
6. **Validate:** Compare the DFT-derived r_{33} against experimental measurements at three conditions (bulk crystal, epitaxial film, cryogenic strained film).

B. S6.2. Before/After Library Fusion

The key diagnostic result from the extraction case study is the ε^∞ uncertainty widening. Adding our PBE value ($\varepsilon_{zz}^\infty = 5.96$) to the existing library (isotropic averages in the range 6.5–7.0) caused the inverse-variance weighted uncertainty to increase by 60%. As discussed in the main text, this widening is a symptom of a tensor-component convention mismatch, not a genuine increase in physical uncertainty. The proper resolution is phase-aware partitioning, which separates zz -component values from isotropic averages before computing weighted means.

The full before/after comparison, including parameter-by-parameter changes and the resulting effect on the derived r_{33} , is generated by running:

```
python scripts/extraction_case_study.py
```

from the paper repository.

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